



CROP YIELD FORECAST

Data Integration, Exploratory Analytics, Machine
Learning Modeling & Prediction Deployment

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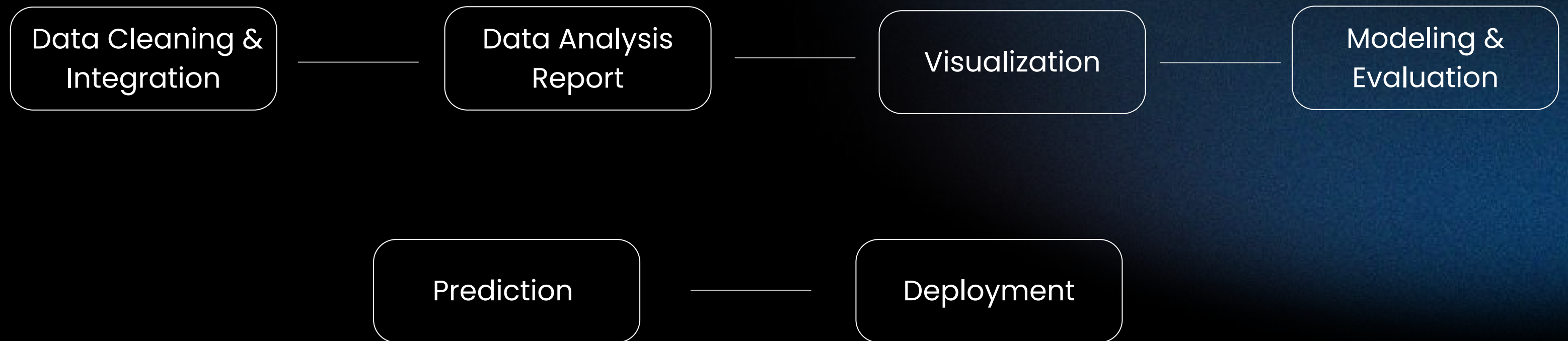
OCTOBER 2026

A close-up photograph of a person's hand holding a green wheat stalk. The hand is positioned in the lower-left foreground, with fingers gently gripping the grain. The wheat stalk is vibrant green and shows the individual grains. The background is a soft-focus field of similar wheat stalks, extending to a distant horizon under a bright, hazy sky. The overall tone is natural and agricultural.

INTRODUCTION

We predicting crop yields through the multi-modal integration of historical production, regional weather dynamics, and commodity market pricing transforms yield estimation from static historical forecasting into an adaptive, risk-aware decision framework.

PROJECT WORKFLOW



Data Cleaning & Integration Pipeline

Raw Sources:

- crop_yield_train.csv, market_prices.csv, regional_weather.csv.

Cleaning Log

- Resolved unit inconsistencies.

Join Map:

- Multi-key join on (region, year, season, crop). Created master_train.csv.

Data Analysis Report

Market Pricing & Revenue Dynamics

- Category Volatility
- Supply Gluts
- Harvest-Window Pricing

Category Yield Patterns

- Yield Ceilings
- Input Sensitivity
- Target Stratification

Agronomic Drivers

- Fertilizer Limits
- Soil & Pest Impact
- Input Bands.

Climatic & Temporal Interactions

- Thermal Stress
- Rainfall Extremes
- Interaction Driver



Modeling & Evaluation

Validation Performance

RMSE: 0.473 t/ha
MAE: 0.349 t/ha
 R^2 : 0.880

Cross-Validation Performance

5-Fold RMSE: 0.488 ± 0.013 t/ha

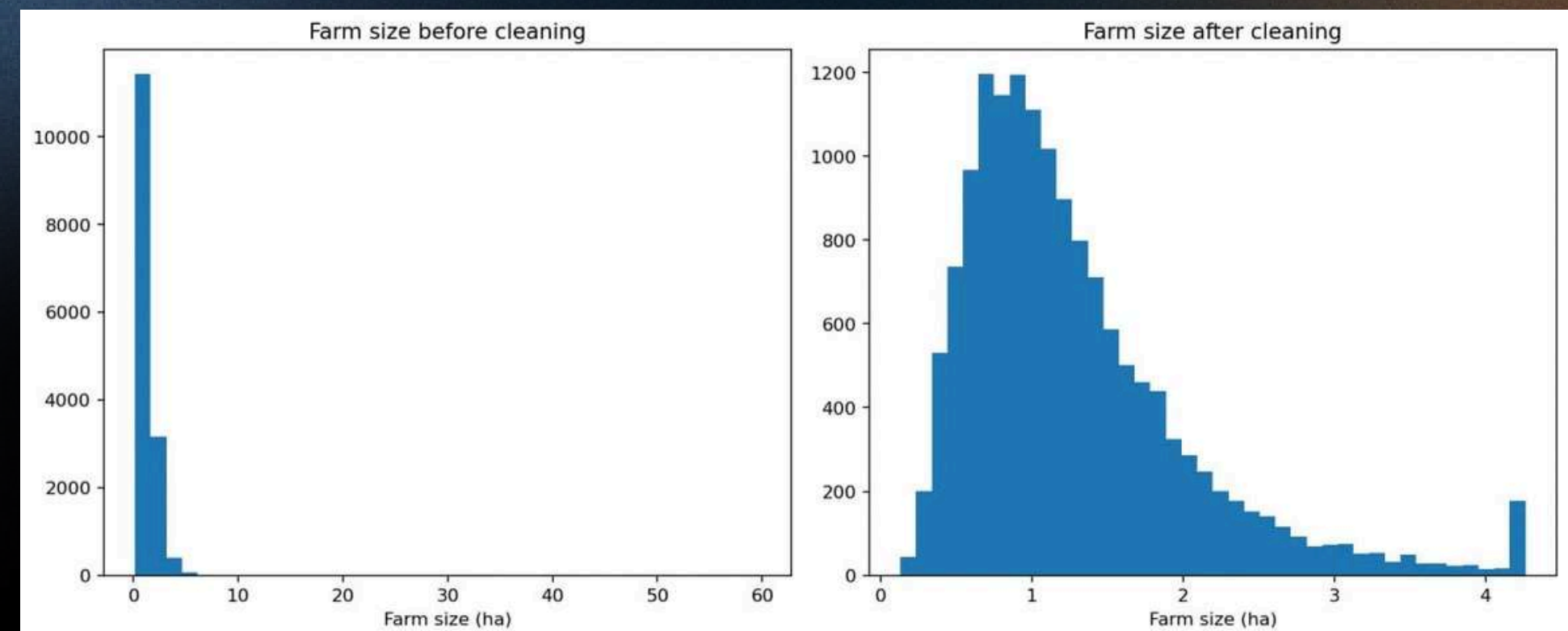
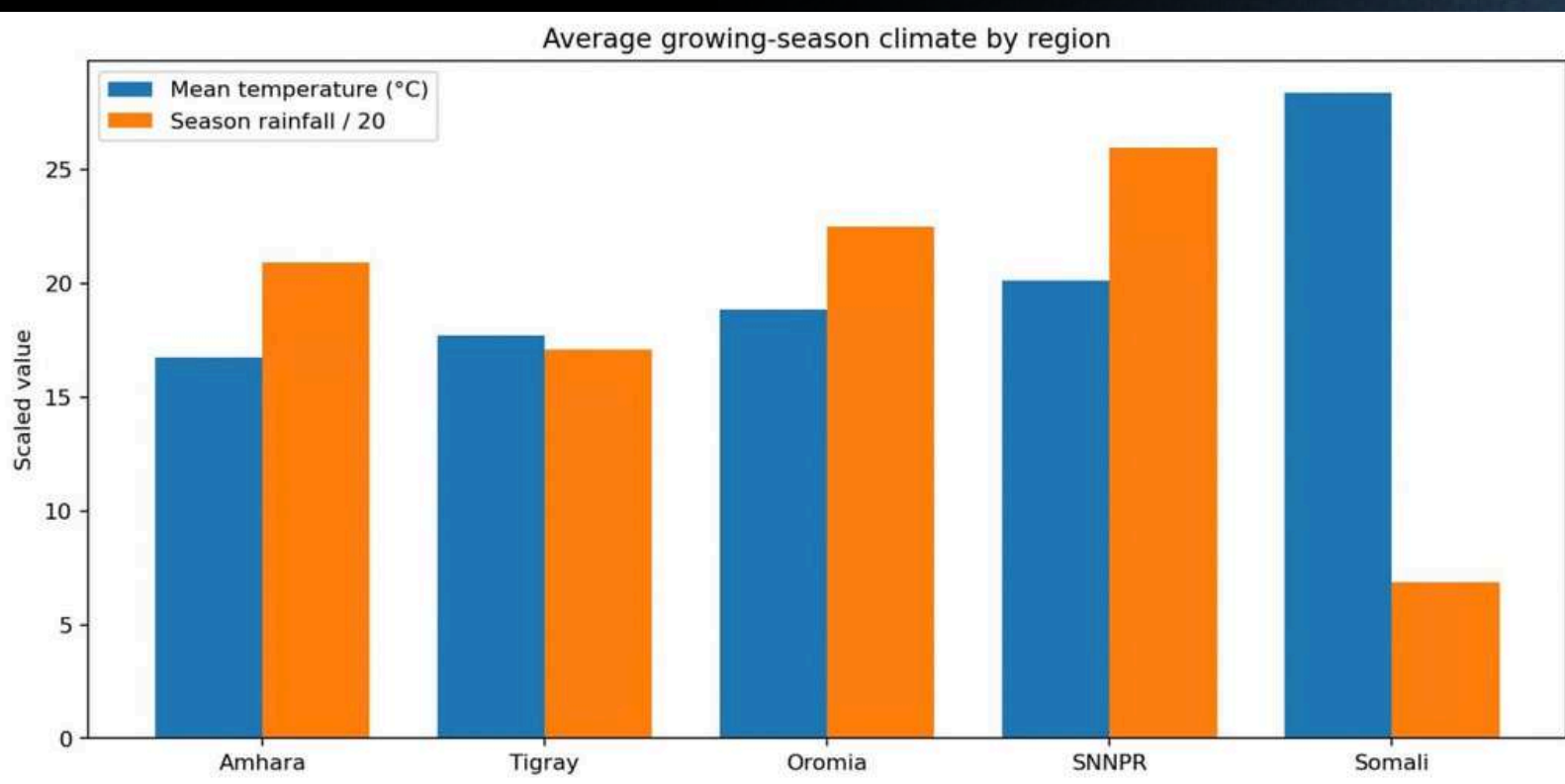
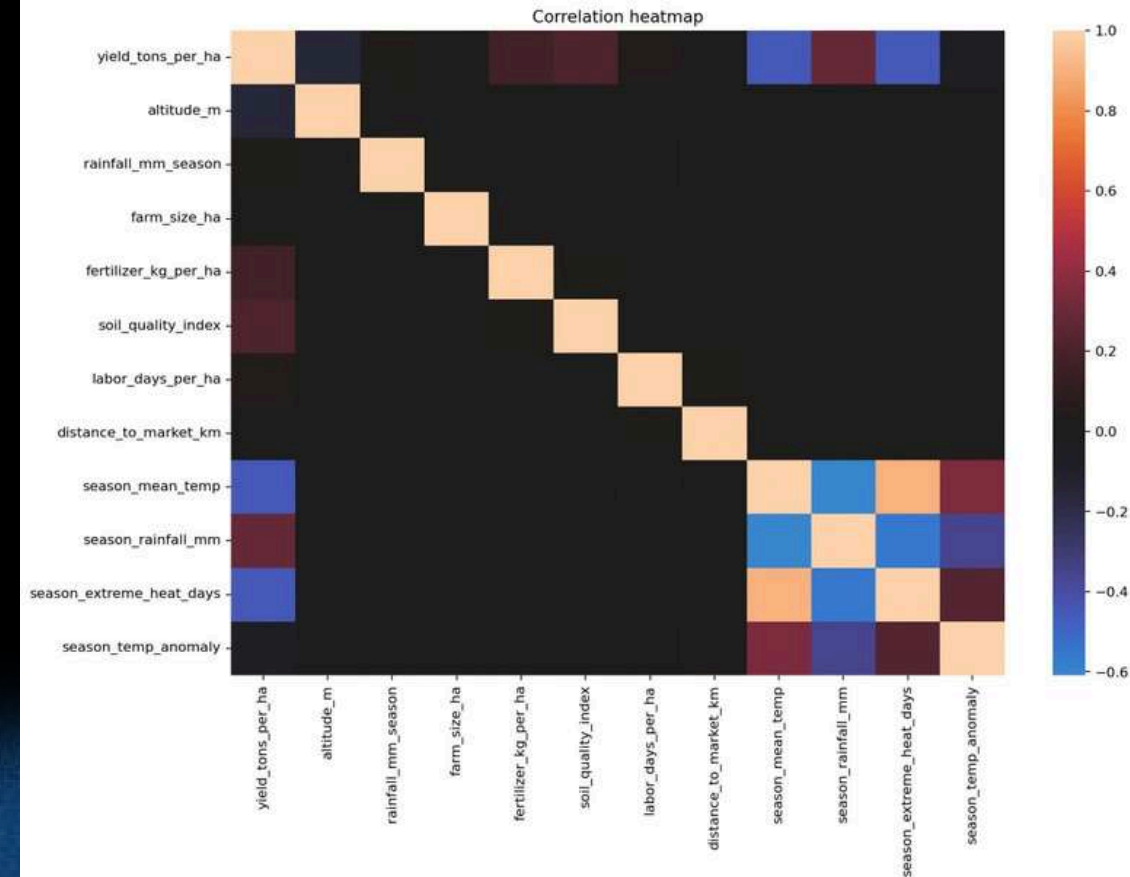
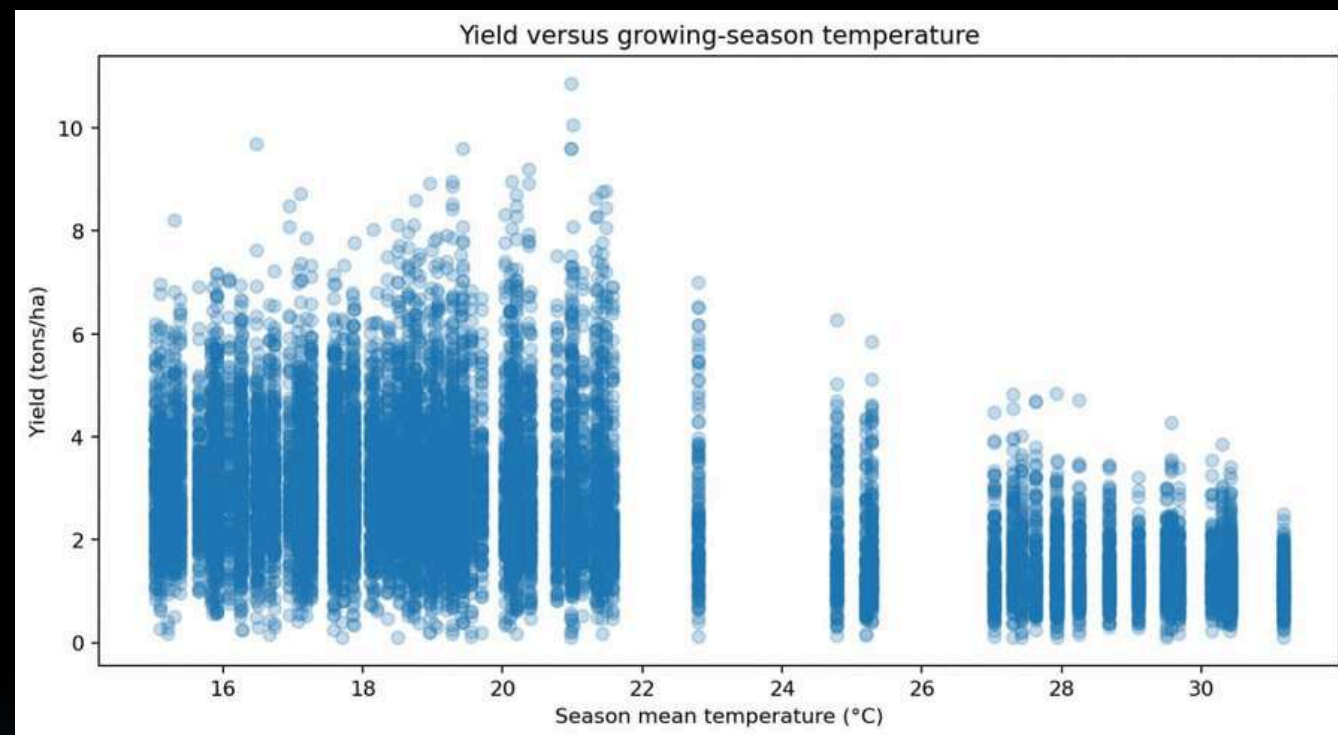
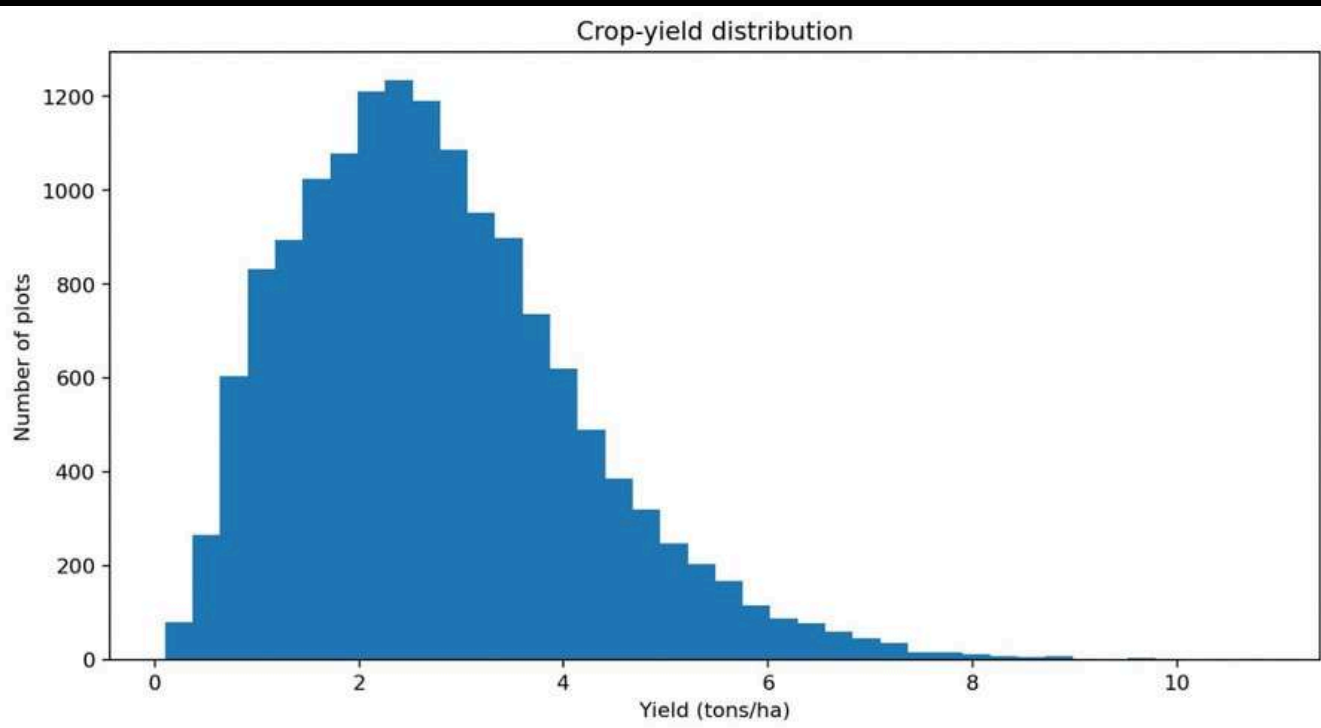
Out-of-Time Testing (2024 Data)

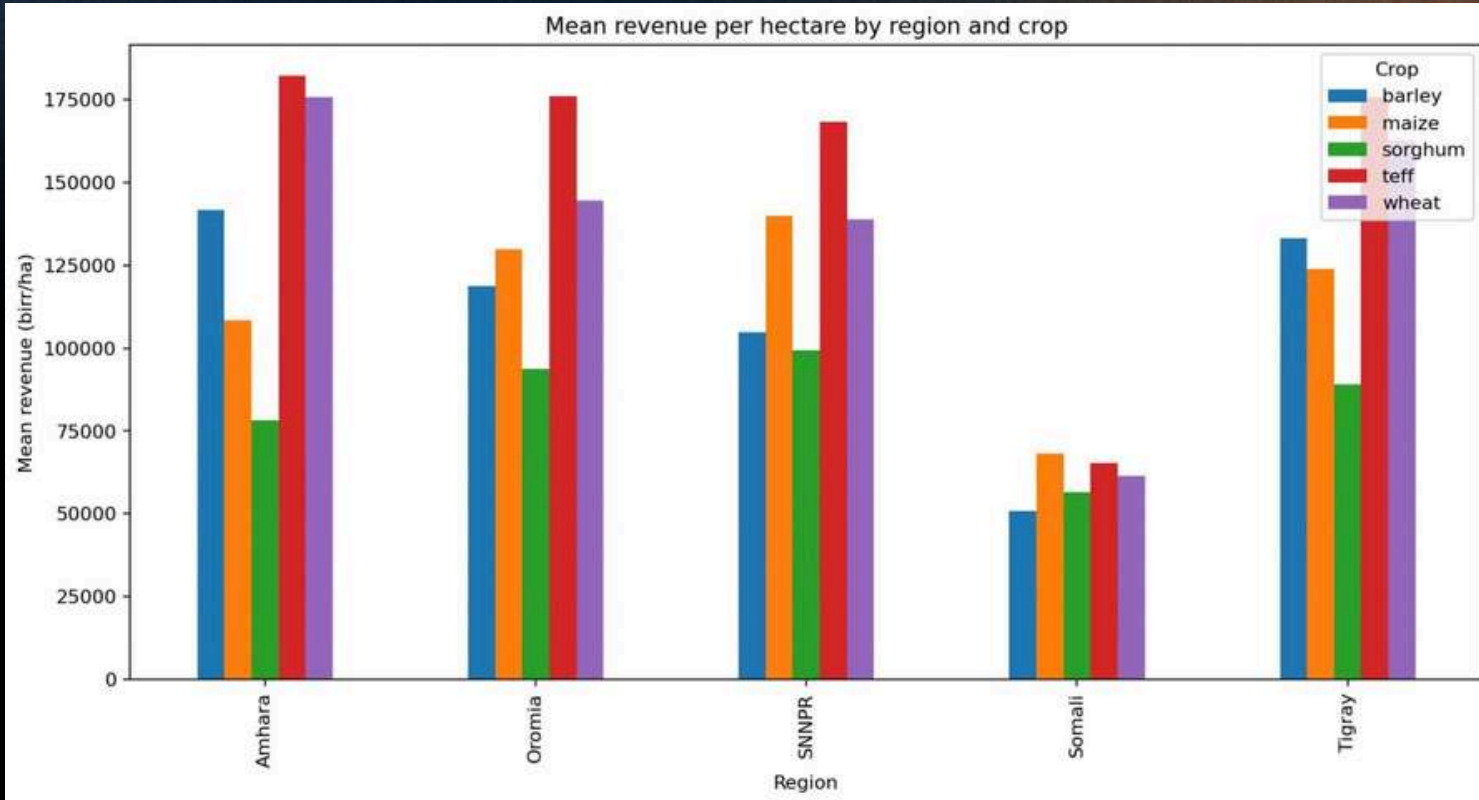
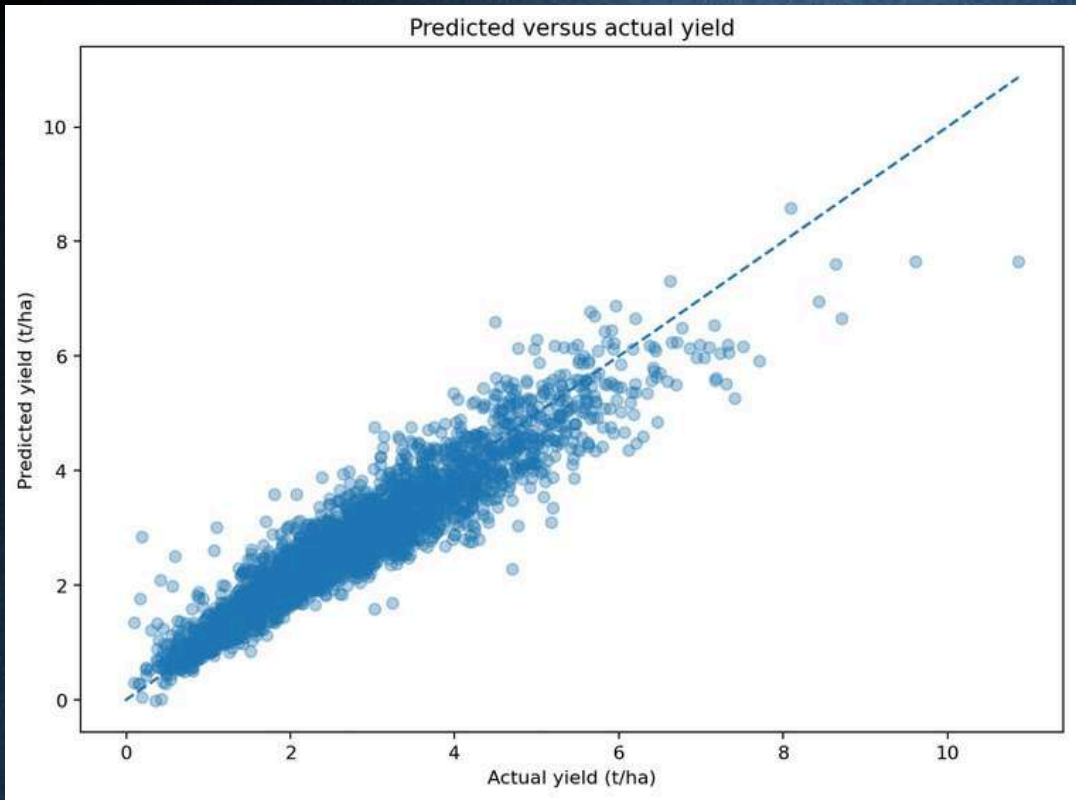
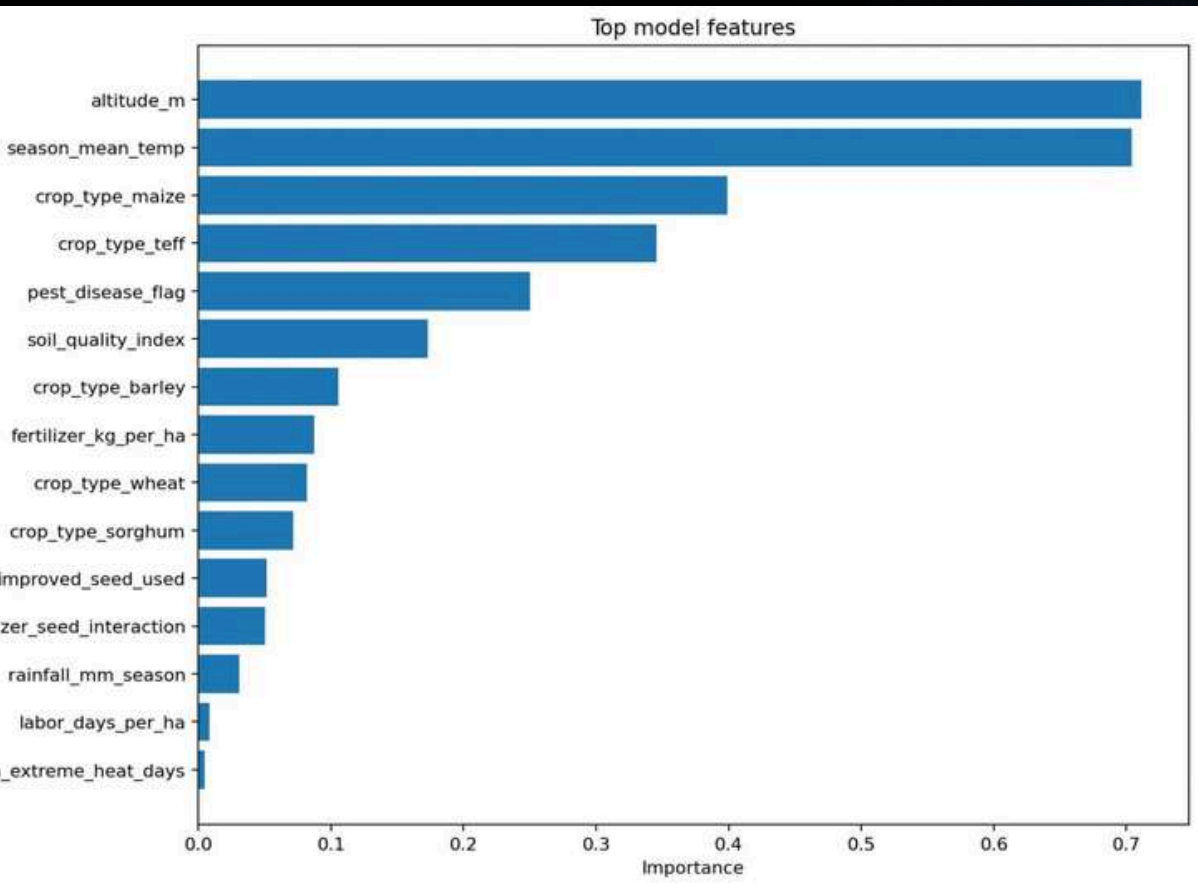
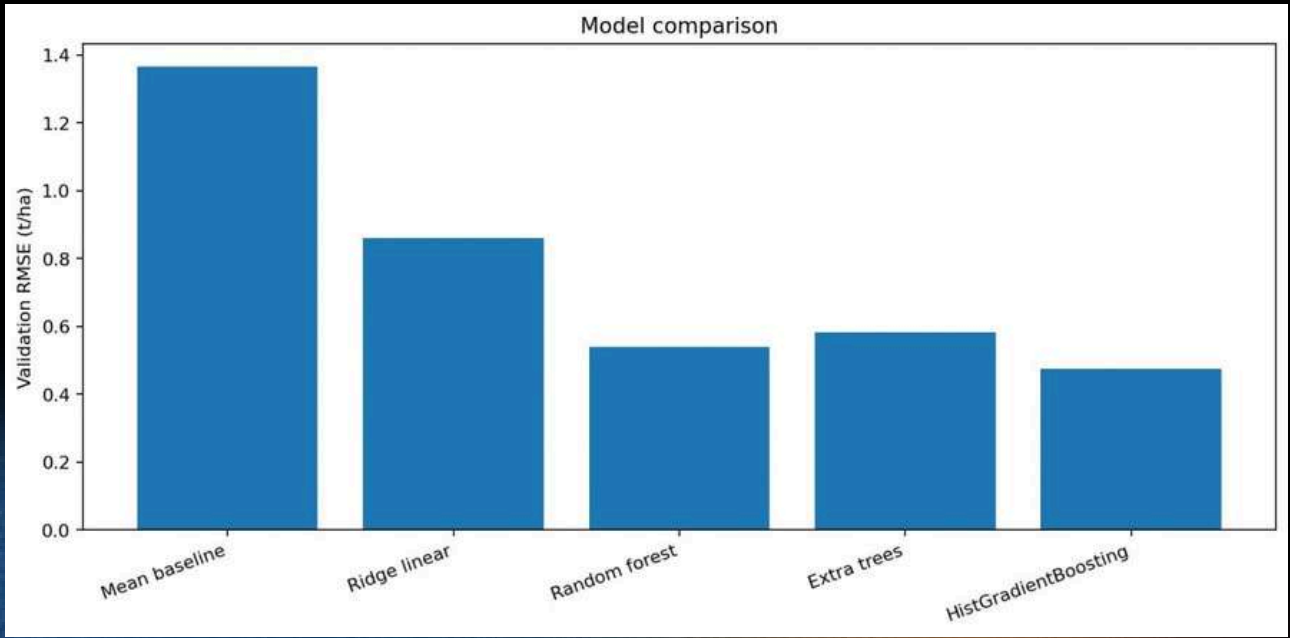
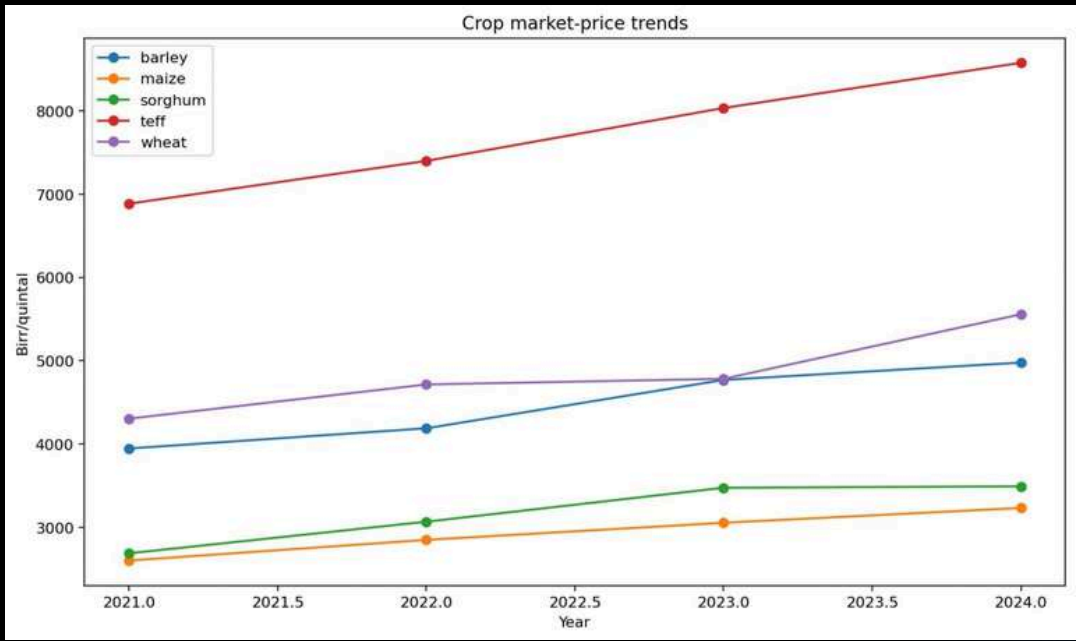
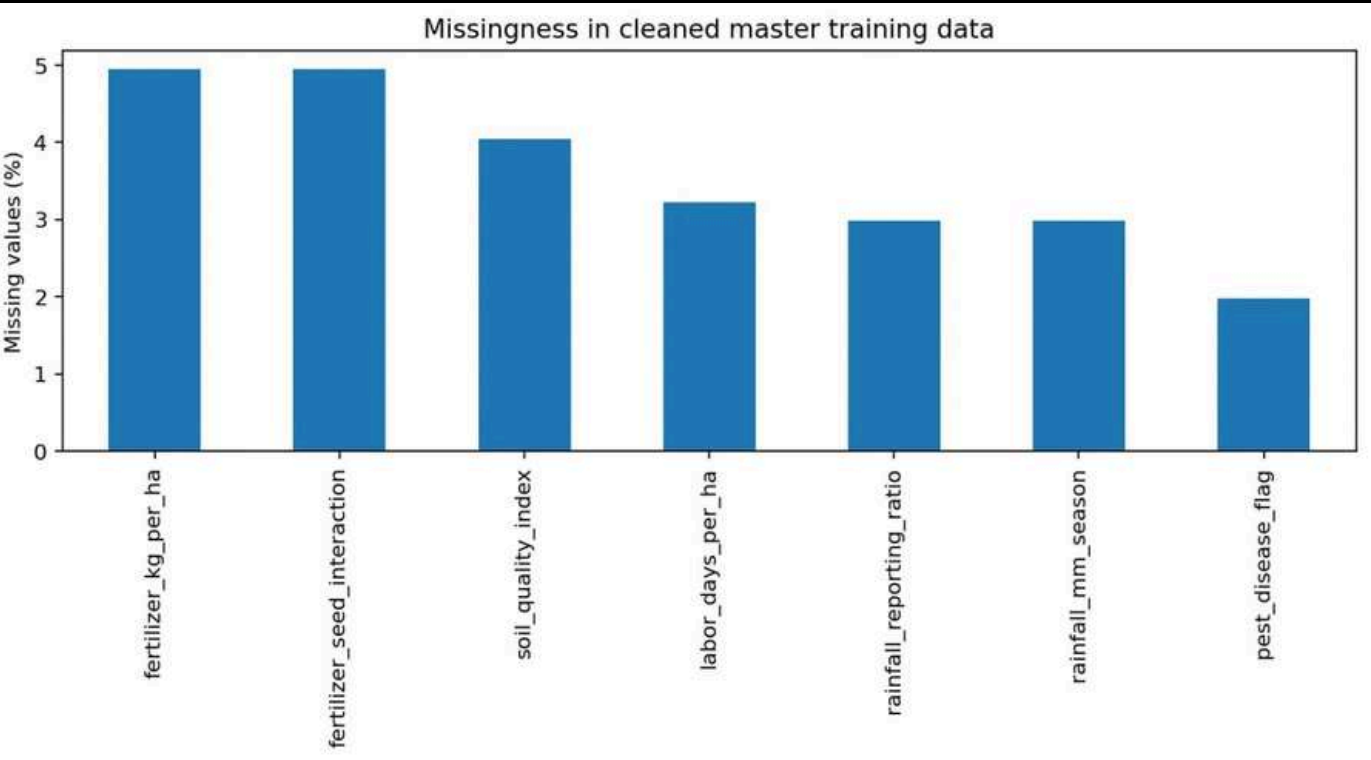
RMSE: 0.506 t/ha

Prediction

The tuned HistGradientBoosting model predicted yields ranging from 1.23 to 6.34 t/ha. PLOT-0155 showed the highest yield potential (6.34 t/ha), while most plots were predicted to produce between 2.5 and 3.5 t/ha, indicating generally stable productivity across the study area. These predictions support yield forecasting and targeted management decisions.

Plot ID	Predicted Yield (t/ha)
PLOT-0172	3.43
PLOT-0172	2.54
PLOT-0161	2.30
PLOT-0186	3.10
PLOT-0151	2.68
PLOT-0163	2.96
PLOT-0179	1.69
PLOT-0164	2.48





LIVE DEMO

127.0.0.1:7860

Ethiopian Smallholder Crop-Yield Forecast

Region

Amhara

Crop type

barley

Year

2024

Planting month

Jun

Altitude (m)

1500

Farm size (ha)

1

Fertilizer (kg/ha)

40

Soil quality (0–1)

0.6

Improved seed

0

1

Pest/disease flag

0

1

Labor days/ha

40

Distance to market (km)

10

Predict yield

Prediction

Predicted yield: 1.79 tons/ha

Estimated revenue: 87,022 birr

Looked up: 15.8 °C seasonal mean, 468.3 mm seasonal rainfall, 4,863 birr/quintal price

Reference region × crop mean: 3.19 tons/ha